Laboratory 02: RF Propagation Modeling with Python

Course: ITAI 4371 – Telecommunications Systems and Networks

Laboratory: Laboratory 02 – RF Propagation Modeling with Python

Student: Brandie Griffin

Professor: Dr. Tawanda Chiyangwa

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Introduction

In this lab, I used Python to simulate Free-Space Path Loss (FSPL). Free-Space Path Loss shows how much a wireless signal weakens as the distance between the transmitter and receiver increases. This concept is important in wireless communications because signals lose strength as they travel farther from the source.

The lab uses a frequency of 2.4 GHz, which is commonly used by WiFi networks. The Python program calculates the path loss for distances ranging from 0.1 kilometers to 20 kilometers and displays the results on a graph. By analyzing the graph, I can observe how signal attenuation increases with distance.

Objective

The objective of this laboratory is to simulate Free-Space Path Loss (FSPL) using Python and visualize how signal loss changes as distance increases. This exercise demonstrates RF signal propagation and helps explain the relationship between distance and wireless signal strength.

FSPL Formula

Free-Space Path Loss is calculated using the following formula:

$FSPL(dB) = 20 \log_{10}(d) + 20 \log_{10}(f) + 32.44$

Where:

d = distance in kilometers (km) f = frequency in megahertz (MHz) FSPL = Free-Space Path Loss measured in decibels (dB)

This formula estimates the amount of signal attenuation that occurs when a radio signal travels through free space.

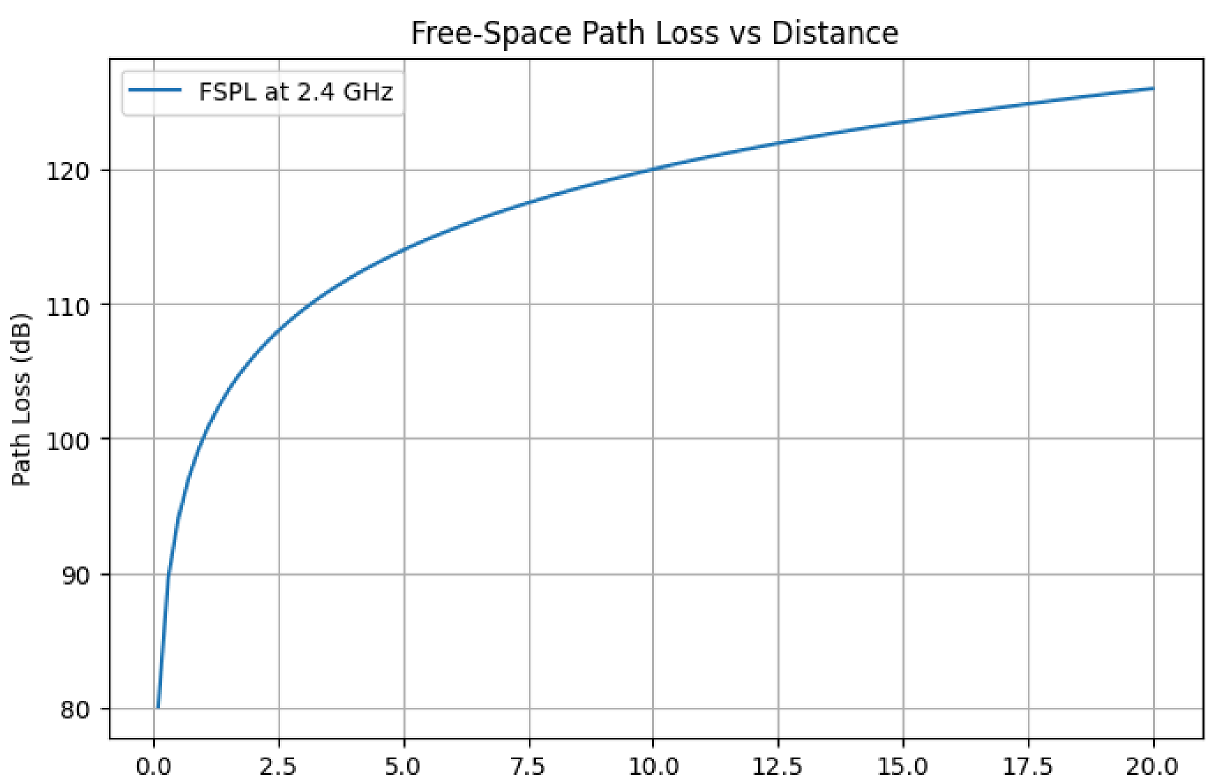
The program imports the NumPy and Matplotlib libraries. NumPy is used to generate distance values and perform mathematical calculations, while Matplotlib is used to create graphs. The frequency is set to 2400 MHz, representing a 2.4 GHz WiFi signal. The program calculates the Free-Space Path Loss over a range of distances and then plots the results to show how signal attenuation increases as distance increases.

```
import numpy as np
import matplotlib.pyplot as plt

# Parameters
frequency = 2400
distances = np.linspace(0.1, 20, 100)

# FSPL calculation
fspl = 20*np.log10(distances) + 20*np.log10(frequency) + 32.44

# Plot
plt.figure(figsize=(8,5))
plt.plot(distances, fspl, label="FSPL at 2.4 GHz")
plt.xlabel("Distance (km)")
plt.ylabel("Path Loss (dB)")
plt.title("Free-Space Path Loss vs Distance")
plt.grid(True)
plt.legend()
plt.show()
```



Output Graph

The graph above illustrates the relationship between distance and Free-Space Path Loss at a frequency of 2.4 GHz. As the distance increases, the path loss also increases, indicating a reduction in signal strength.

Discussion of Results

The graph shows a steady increase in Free-Space Path Loss as distance increases. At shorter distances, the signal experiences less attenuation, while at greater distances the signal weakens significantly. This demonstrates the importance of considering distance when designing wireless communication systems.

The results confirm that wireless signals become weaker as they travel farther from the transmitter. Engineers use these calculations when planning wireless networks to ensure adequate coverage and signal quality.

Conclusion

After running the program, I learned that Free-Space Path Loss increases as distance increases. This means that the farther a wireless signal travels, the weaker it becomes. The graph clearly shows that path loss is lower at shorter distances and higher at longer distances.

This lab helped me understand why wireless networks require strong transmitters, antennas, or repeaters when signals must travel over long distances. Understanding FSPL is important for designing reliable wireless communication systems.